Newumal Weerasinghe

newumaalweerasinghe@gmail.com +94 71 7349697 linkedin.com/in/newumal Homagama, Colombo, Sri Lanka
github.com/newumal youtube.com/@buildwithnewumal

PROFESSIONAL SUMMARY

Associate Tech Lead and Team Lead with 7+ years of experience building and leading SaaS products, and an early hands-on AI adopter with 2+ years applying AI across the full engineering lifecycle — design, product thinking, and delivery — while keeping decisions human-owned. Proven track record leading AI initiatives (LLMs, RAG, agents, Copilot), cost optimization, and root-cause-driven performance improvement on cloud-native AWS architectures and a full-stack of React.js, Node.js, and PostgreSQL. Bridges product and engineering — turning business goals into scalable, high-quality delivery while mentoring engineers and driving team performance.

WORK EXPERIENCE

Associate Tech Lead & Team Lead Jan 2026 – Present
Velaris | Moratuwa, Sri Lanka

- Lead a cross-functional engineering team — owning sprint planning, technical direction, code reviews, and engineer mentoring.
- Proposed and led a new AI feature initiative: built a proof-of-concept praised by the CEO, Head of AI, and product owners, shipped it to production after user feedback and fine-tuning, and scaled the approach across teams in partnership with the AI team — directly supporting new-customer onboarding and sales.
- Introduced a data-driven prioritization process to tackle churn and reliability: analyzed the priority-issues channel, produced root-cause reports with permanent fixes, dev/QA effort estimates, and timelines, and aligned the product owner and engineering lead to act on them. Shifted the team from patching symptoms to permanent solutions, unlocking capacity for new features. Later adopted by other teams.
- Drove system stability, cost, and scalability initiatives with cost- and product-thinking — including customer 360-view improvements, a major platform version (v4) rollout, and scaling to 500K+ record support.
- Stepped up as acting team lead during a senior lead's absence — owned Scrum ceremonies, divided work across engineers, coordinated with QA and PM, and delivered multiple production releases under pressure while maintaining stability.
- Mentored several engineers across teams (integration work, Slack/ticketing, testing strategy) and contributed to hiring through engineering interviews.
- Championed responsible AI adoption across the team — ran enablement sessions (at the engineering lead's and product owner's request) on applying AI across the engineering lifecycle without losing engineering judgment, introduced an AI best-practices framework and standards, and built custom agents for high-quality, low-hallucination implementation; most of the team now uses AI effectively in daily work.
- Personally delivered features on time and defect-free over 2+ years of AI-assisted development, using AI for design, brainstorming, trade-off negotiation, and repetitive work while owning all engineering decisions.

Senior Software Engineer (Full Stack) Jul 2022 – Jan 2026
Velaris | Moratuwa, Sri Lanka

- Built an AI-powered Copilot feature enabling users to generate charts and visualizations (line, pie, bar, KPI) from natural language queries and create interactive Canvas dashboards; improved Copilot context so users can ask richer questions.
- Engineered user-facing AI features hands-on — working across multiple LLMs, RAG, context graphs, and agentic systems, with a focus on LLM cost management.
- Led user-integration projects end-to-end across Google Cloud and Microsoft Azure — building and verifying OAuth apps (app registration, consent screens, Google/Microsoft app-verification review, and auth/token flows) for Gmail, Outlook, calendars, and meeting transcripts.

- Delivered AI meeting-intelligence integrations — Glyphic AI and Microsoft Teams meeting transcripts (Teams shipped with zero post-release issues under a tight deadline) — plus Google/Outlook calendars and Gmail, via new microservice pipelines and frontend integration.
- Architected and delivered a Salesforce integration suite — 5 components delivered against 3 planned (Notes sync and pushback, Opportunity/Risk sync and pushback) — including a streaming-data design to handle large datasets with minimal defects.
- Re-architected core email processing (including a global event-handler fix to drop unwanted events during initial sync), delivering significant performance and cost improvements with zero post-release bugs.
- Led AWS cost-optimization — right-sized resource usage and rationalized logging to cut operational costs across compute, I/O, and database layers; designed a token health monitoring system with early revocation detection and proactive alerting.
- Built database infrastructure for Mixpanel integration enabling cross-team analytics; re-architected and delivered major performance gains to the customer 360 activity feed via pagination and a data migration with zero data loss.
- Created a reusable Git integration template standardizing Teams/Email/Calendar patterns, accelerating consistent user-level integrations across the team.
- Developed SaaS features using React.js, Node.js (NestJS / Express), Redux, PostgreSQL, and AWS (Lambda, EC2, S3, SQS, Step Functions, CloudWatch, CloudFormation); maintained strong test coverage with Jest and close QA collaboration.

Software Engineer (Full Stack) Mar 2021 – Jul 2022

Sezenta (Pvt.) Ltd | Battaramulla, Sri Lanka

- Developed web applications using React.js, Node.js, MongoDB, and AWS.
- Built mobile applications with React Native and Node.js for cross-platform functionality.
- Built REST APIs using Node.js, Express, and NestJS frameworks.
- Leveraged AWS services to enhance application scalability and performance.

Development Engineer & Co-Founder Feb 2019 – Mar 2021

STEM Labs (Pvt.) Ltd | Nugegoda, Sri Lanka

- Delivered IT and STEM education lectures covering robotics and electronics.
- Led web design and development projects using PHP, JavaScript, CSS, and WordPress.
- Developed software and hardware solutions, including a patented IoT hardware product.

Associate Software Engineer Feb 2017 – Feb 2019

Effective Solutions (Pvt.) Ltd | Moratuwa, Sri Lanka

- Built software using the Symfony Framework, including a loyalty platform and student enrollment system for Trinity College.
- Developed e-commerce websites for clients including Softlogic and TyreOnline.
- Designed IoT platforms using Arduino, MQTT, Node.js, and Node-RED; integrated GPS tracking via REST API.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, TypeScript, Node.js, React.js, NestJS, Express.js, Redux, React Native, PHP, Python (learning)

Databases: PostgreSQL, MongoDB, MySQL, DynamoDB, TypeORM

Cloud & DevOps: AWS (Lambda, EC2, S3, SQS, Step Functions, CloudWatch, CloudFormation), Google Cloud Platform, Microsoft Azure, CI/CD, Docker

Identity & Auth: OAuth 2.0, Google & Azure app registration and verification, consent screens, SSO, token management

AI Engineering: LLM integration, RAG, context graphs, custom AI agents, prompt engineering, evals, LLM cost management, AI-assisted SDLC — GitHub Copilot, Cursor, Claude Code

Integrations & APIs: REST APIs, third-party integrations (Google, Outlook, Gmail, Salesforce, Glyphic AI, Mixpanel)

Testing & Tools: Jest, Git, JIRA, Confluence

Leadership: Technical leadership, team management, agile delivery, system design, code review, mentoring

EDUCATION

BSc in Electronics & IT
University of Colombo | Colombo, Sri Lanka

ACHIEVEMENTS

Patented IoT hardware product: Co-designed and shipped a patented IoT hardware product as co-founder of STEM Labs.

Technical content creator: Publish engineering and technical-leadership content on YouTube (@buildwithnewumal).

Kotiyak Watina Adahasak — Season 2, Final 24 (2018): Reached the final 24 of a national innovative-ideas competition broadcast on Sirasa TV, Sri Lanka.